MBA · Executive Program · Integrated Lecture

Fairness in Financial AI

Definition · Measurement · Mitigation · Governance

Algorithmic Fairness in Financial AI — From Unawareness to Counterfactual Fairness

Edition II — with added intuitive examples and visuals.
What "fair" means in lending, credit, insurance and product recommendation, how to measure and fix it, and who is accountable.
Reflects regulation as of October 2026 (EU AI Act delay, Korea's AI Basic Act in force, US Reg B amendment).

This lecture follows three claims

Fairness is not a single number but a set of criteria you must choose from

1

Removing protected attributes does not remove discrimination

Proxies such as postcode, school and name bring the attribute back. Measure discrimination in outcomes, not in the variable list.

2

Fairness definitions conflict, so a choice is required

With different base rates, calibration and equal error rates cannot hold together. The harm you want to prevent sets the definition.

3

Measurement, mitigation and audit last only inside the approval process

Put group-level metrics in the model card and set gates. 2026 regulation is moving toward requiring this record.

Today's question: "For whom, which errors, and how equal?"

01

PART 1 · Fundamentals

The Structure of the Fairness Problem

Definition → confusion-matrix metrics → sources of bias → landmark cases

Algorithmic fairness deals with outcome gaps between groups

Algorithmic Fairness — premise: bias arises from data and usage, not the model itself

**If people with the same qualifications receive different decisions because of their group,
the model is not fair, however accurate it is.**

Why it matters especially in finance

- Loans, limits and premiums directly change life opportunities
- Past discrimination remains in the training data
- Automated decisions replicate a single bias at scale

Terminology

- Protected attribute A: gender, race, age and other prohibited grounds
- Label Y: actual outcome (repaid=1, default=0)
- Prediction $\hat{Y}$: model decision (approve=1, reject=0)

Every fairness metric comes from three ratios of the confusion matrix

TP=repayer approved, FN=repayer rejected, FP=defaulter approved, TN=defaulter rejected

$$\underbrace{\text{TPR} = \frac{TP}{TP + FN}}_{\text{approval rate among the qualified}} \qquad \underbrace{\text{FPR} = \frac{FP}{FP + TN}}_{\text{approval rate among the unqualified}} \qquad \underbrace{\text{PPV} = \frac{TP}{TP + FP}}_{\text{repayment rate among the approved}}$$

	Approve ($\hat{Y}$=1)	Reject ($\hat{Y}$=0)
Actually repays (Y=1)	TP — given a chance	FN — denied a chance
Actually defaults (Y=0)	FP — caused a loss	TN — correctly screened out

Whose harm is each cell?

- FN harms the applicant
- FP is the lender's loss
- Which cell to equalize separates the definitions

The two kinds of error are paid for by different people

Placing a fire alarm next to loan screening shows the weight of FP vs FN

	Fire alarm	Loan screening
FP (false positive)	Rings with no fire — a wasted trip	Approves someone who cannot repay — lender loss
FN (false negative)	Silent during a fire — severe damage	Rejects someone who would repay — applicant loses a chance
Who bears the cost	All residents	FP: the lender; FN: the individual applicant

Key point

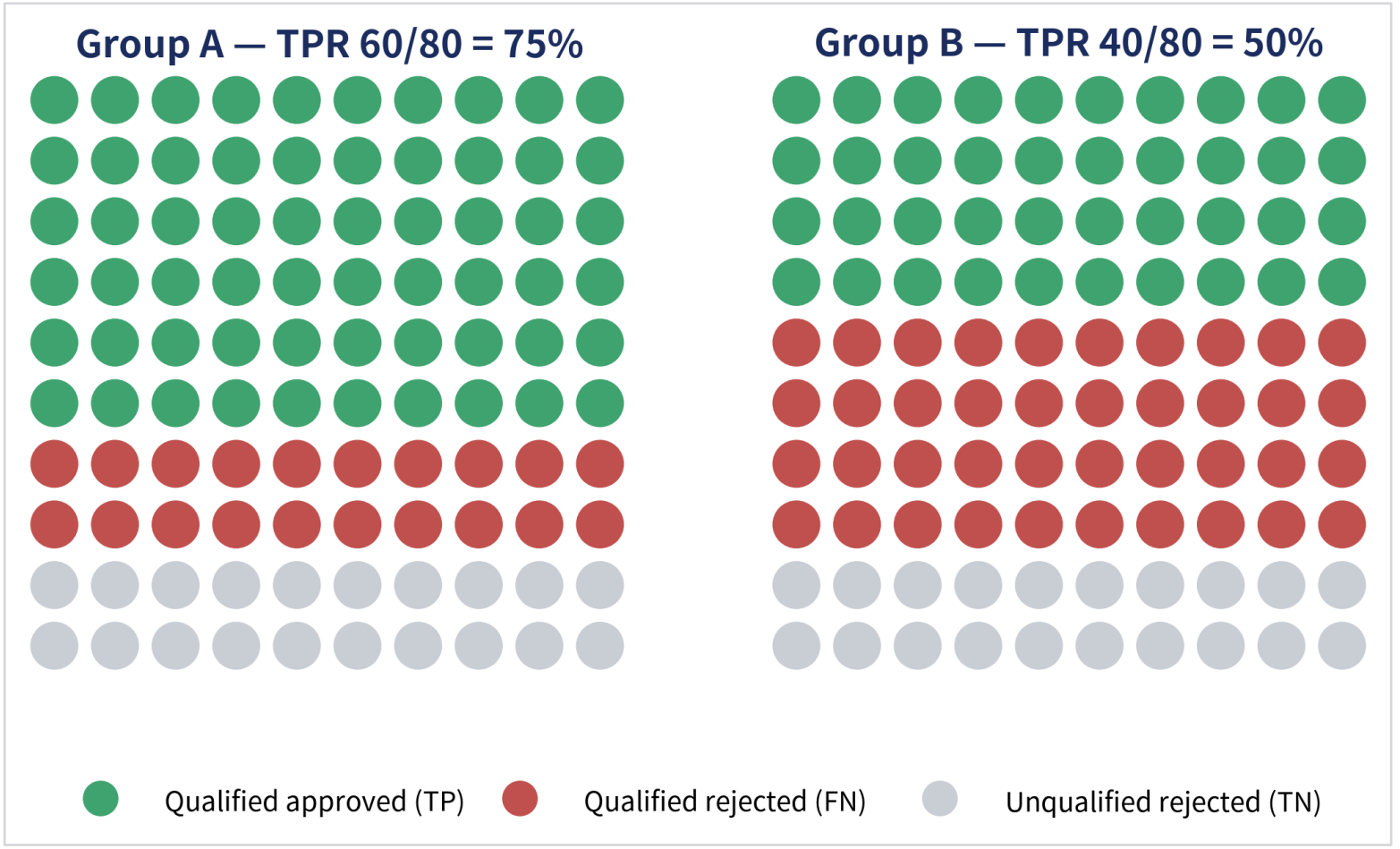
- Look where FNs concentrate for equal opportunity; add FPs for equalized odds.

Drawn as 100 dots, unequal treatment of equal qualifications shows

Two groups of 100 each, with the same 80 actual repayers (FML Week 14 example)

How to read

- One dot = one person
- Green = given a chance
- Red = denied a chance
- B has twice the red



Bias enters at four stages of data generation

Suresh & Guttag (2021) taxonomy, adapted to financial cases

Source	What happens	Financial example
Historical bias	Past discrimination remains in the labels	Past loan rejections are learned as ground truth
Representation bias	Some groups have few samples	Little data on young people and immigrants with short credit histories
Measurement bias	The same trait is measured differently	Alternative data covers only some groups
Aggregation bias	One model is used for everyone	Salaried-worker model applied to self-employed with different income structures
Proxy	Neutral variables stand in for the attribute	Postcode, school and name reveal race and gender

Five landmark cases show the axes of the fairness debate

Updated with the latest findings — Apple Card was found to have no violation by the 2021 regulator investigation

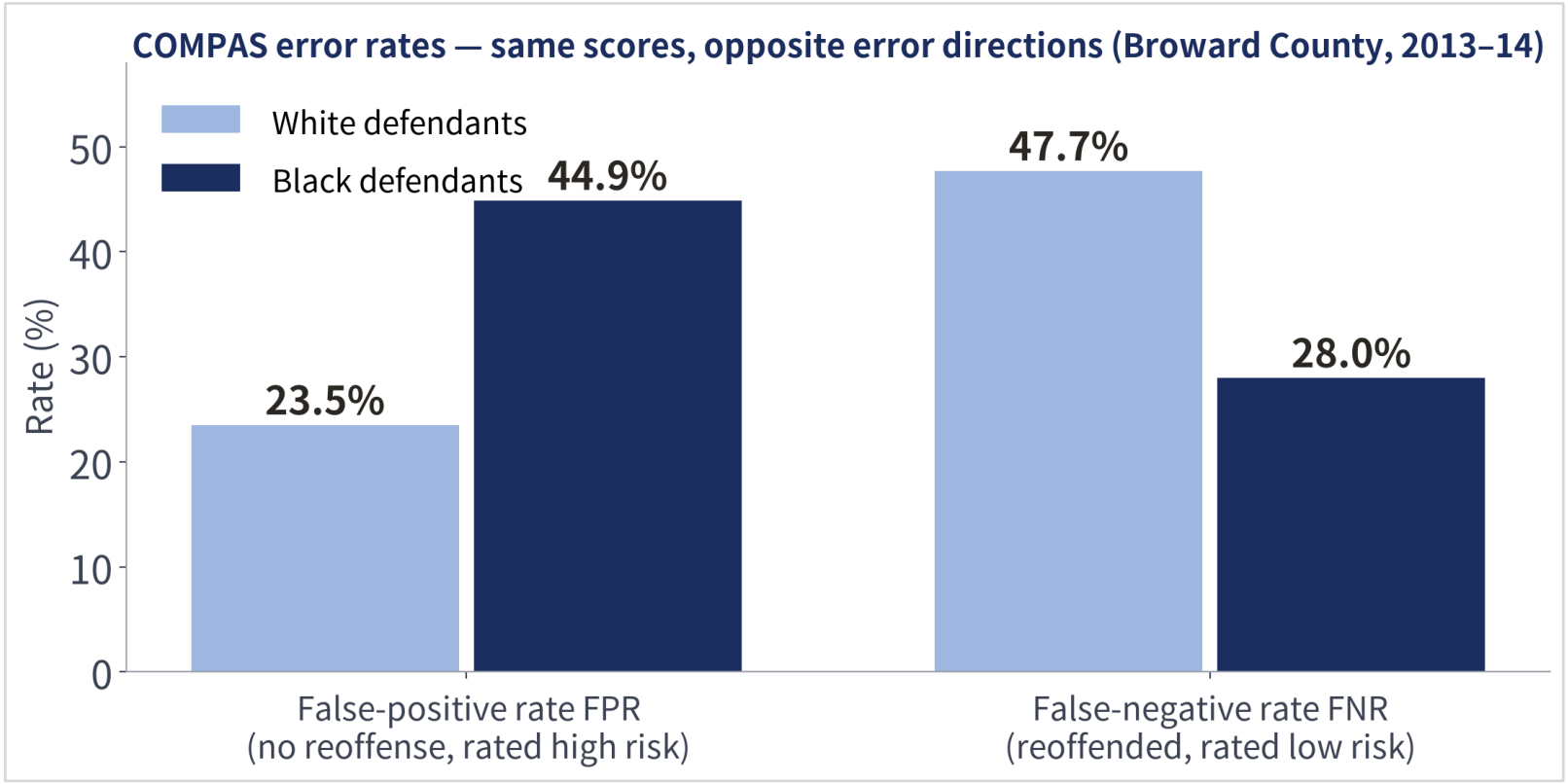
Cases	What went wrong	Lesson
Amazon hiring (2014–18)	Trained on mostly male résumés; penalized the word 'women's'	Deleting variables leaves the pattern
COMPAS (2016)	False-positive rate for Black defendants 45% vs White 23%	Error rates and calibration conflict
Apple Card (2019–21)	Spousal limit gap controversy, no gender variable	No violation — lack of explanation cost trust
Earnest (2025)	School-level default-rate variable had disparate impact	Using proxies without checks invites sanctions
LLM mortgage experiment (2024–25)	Same documents, only race changed, gave different approvals	General-purpose LLMs must be audited too

COMPAS had equal accuracy but errors in different directions

ProPublica tracked two-year recidivism of 10,000+ defendants in Broward County, Florida

How to read

- Accuracy 59% vs 63%
- Black: over-labeled high risk
- White: under-labeled low risk
- Scores themselves calibrated



The two sides were using different fairness definitions

ProPublica used equal error rates; the developer Northpointe used score calibration

ProPublica's claim

- Black non-reoffenders are twice as likely to be labeled high risk
- FPR and FNR differ by group, so it is unfair

Northpointe's rebuttal

- At the same score, the recidivism rate is the same regardless of race
- Scores are calibrated, so it is fair

Both are numerically right. If recidivism base rates differ by group, the two criteria cannot hold together (→ impossibility theorem).

The Apple Card probe found a lack of explanation, not a violation

Social-media complaint in Nov 2019 → conclusion by the New York State Department of Financial Services (NYDFS) in Mar 2021

20×

Spousal limit gap claimed by the complainant

~400,000

New York applicants analyzed by NYDFS

No violation

Neither intentional gender discrimination nor disparate impact found

What set the limits

- Each spouse's own credit score, debt and income
- Jointly held assets were not counted

Lessons for this lecture

- Suspicion arises even without a gender variable
- Reasons for rejections and limits must be explainable

The Earnest settlement shows proxy variables invite sanctions

July 2025: $2.5 million settlement between the Massachusetts Attorney General and student lender Earnest

Practice cited	Why it is a problem	Settlement terms
School-level Cohort Default Rate variable	Adverse outcomes for Black and Hispanic applicants	Stop using the variable
Automatic rejection of non-green-card holders (knockout rule)	Risk of national-origin discrimination	Abolish the rule
No disparate-impact testing of models	Operating with unknown risk	Annual fair-lending testing and reporting
Inaccurate adverse-action notices	Applicants cannot learn the reason	AI governance structure and documentation

The company denied the allegations but settled. Sanctions were possible under existing consumer-protection law, without a new AI law.

General-purpose LLMs answered differently by race in loan screening

Bowen · Price · Stein · Yang — 6,000 experimental applications built from 1,000 real 2022 HMDA applications

+120 pts

Extra credit score Black applicants need for the same approval rate (GPT-4 Turbo)

+30 pts

Extra score needed for the same interest rate

≈ 0

Approval gap after instructing "decide without bias"

What it shows

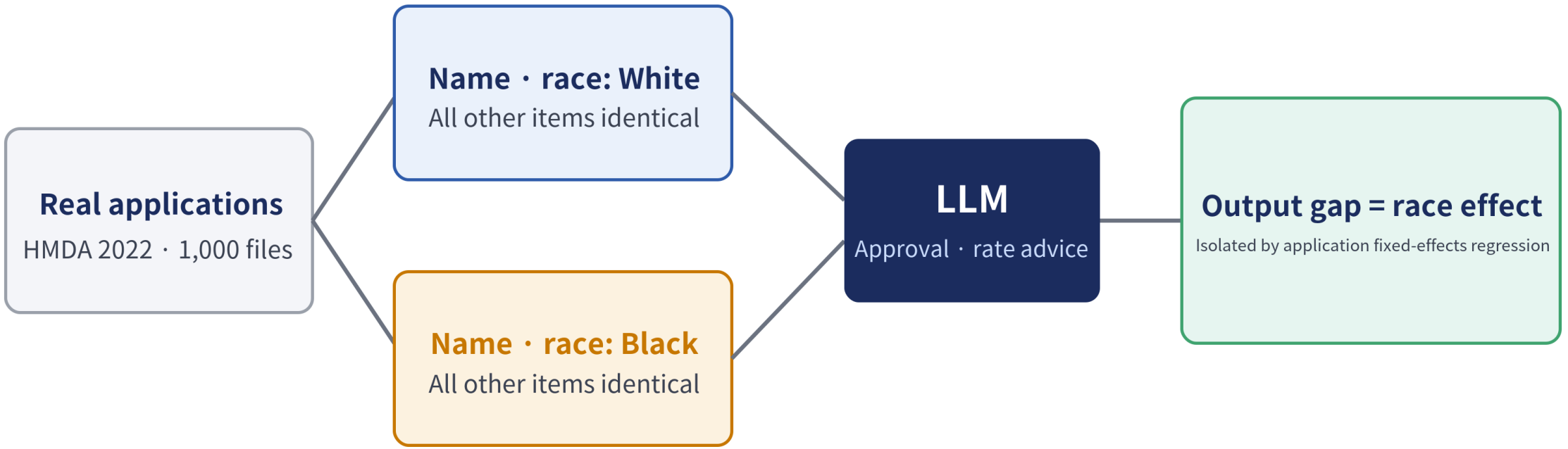
- A name alone conveys a racial signal
- Gaps are larger for low-credit, high-risk applications

Practical implications

- LLM advice and recommendations are also subject to audit
- Prompt mitigation is a complement, not a substitute

Audit design fixes the documents and changes only the racial signal

Make two copies of the same application and compare outputs; any difference comes only from the racial signal



Audit-study design changing only the racial signal on identical documents — lenders can apply it directly to their own LLMs

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PART 2 · Fairness Definitions

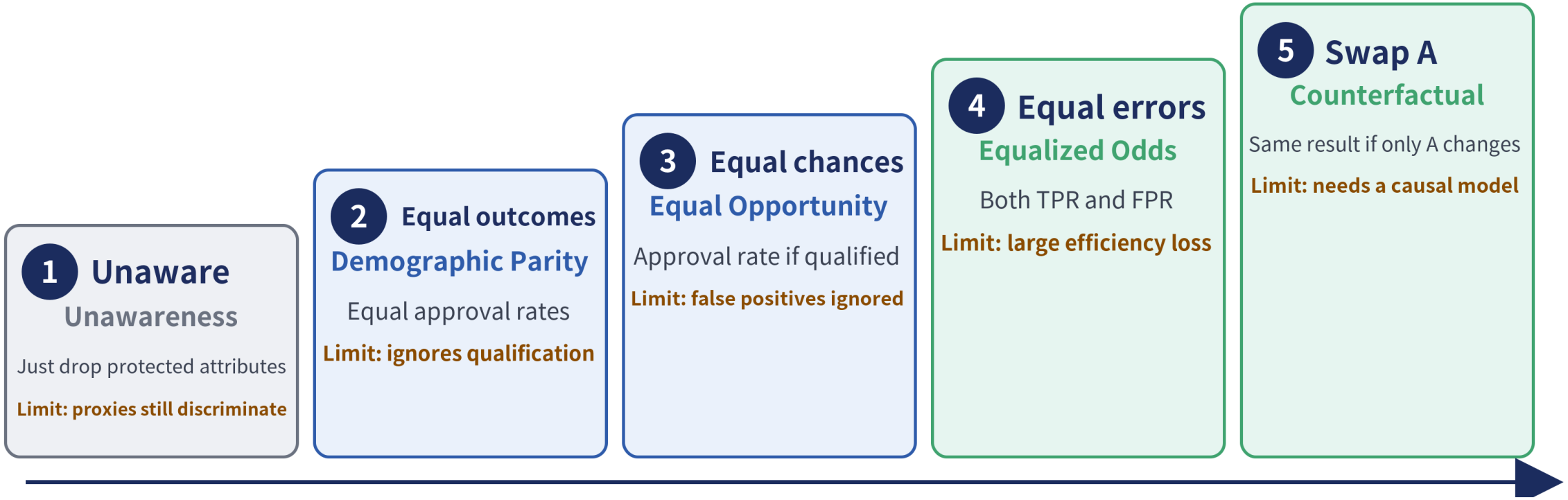
The Five-Rung Fairness Ladder

Unawareness → demographic parity → equal opportunity → equalized odds → counterfactual fairness

Fairness definitions become more refined in five steps

Each step up needs more data (actual outcomes) and causal knowledge (what produces what)

Higher requirements need more data and causal knowledge



Each step answers a different question

The source's height analogy: from "just stand, no measuring" to "same outcome regardless of gender"

Step	Question asked	Philosophical basis	Implementation difficulty
1 Unawareness	Is it fair if protected attributes are not used?	Formal equality	Low
2 Demographic parity	Are approval rates equal across groups?	Equality of outcome	Low
3 Equal opportunity	Are approval rates equal among the qualified?	Meritocracy	Medium
4 Equalized odds	Equal for both qualified and unqualified?	Procedural justice	High
5 Counterfactual fairness	Is the outcome the same if only the attribute changes?	Causal justice	Very high

The source stopped at four steps; counterfactual fairness was added as a separate step, not part of equalized odds.

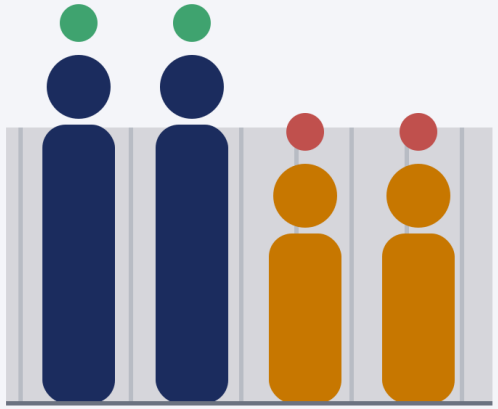
The source's height analogy shows the four steps at a glance

Watching a game over a fence — clearing the fence is 'qualification', seeing the game is the 'benefit'

Navy · orange = two groups, green dot = sees the game (benefit), red dot = cannot see — the source's height analogy

1 Unawareness

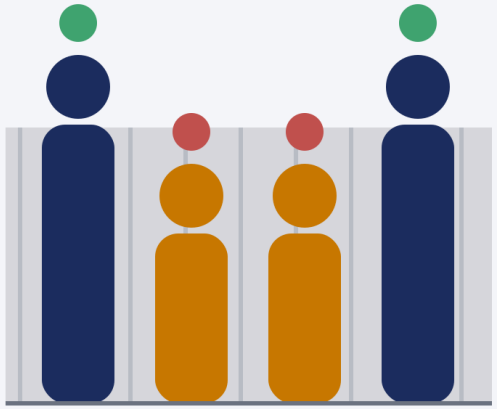
"Don't measure height, just stand"



Short people are blocked by the fence

2 Demographic parity

"Line up equal numbers of men and women"



Ratio matched, but some can't see

3 Equal opportunity

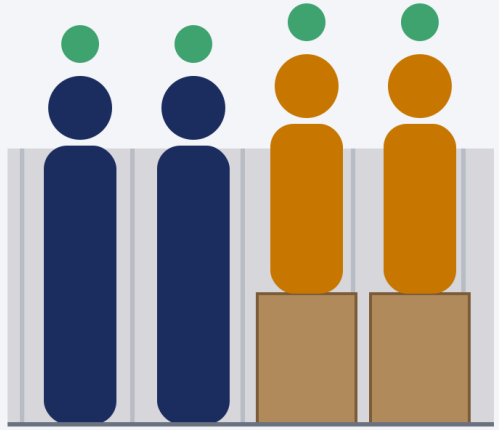
"Half and half among those who can see"



Half and half among those over the fence

4 Equalized odds

"Same result for everyone"



Boxes give everyone the same result

Each definition resembles a familiar everyday rule

In class we build intuition with everyday cases before the formulas

Definition	Everyday analogy	Question in finance
Demographic parity	Split club places equally across departments	Are approval rates equal across groups?
Equal opportunity	Students above the cutoff are picked with equal probability regardless of department	Are approval rates equal for those who would repay?
Equalized odds	Airport screening has equal hit and false-alarm rates regardless of nationality	Are both TPR and FPR equal?
Calibration	'70% chance of rain' means 70% in every neighborhood	Is repayment rate equal at the same score?
Individual fairness	Two students one point apart get similar grades	Do similar customers get similar decisions?
Counterfactual fairness	Résumés differing only in name get the same callbacks	Is the decision the same if only the attribute changes?

Proxies revive discrimination even after protected attributes are deleted

Fairness through Unawareness — only removes A from the input; does not ensure independence of outcome ($\hat{Y} \perp A$)

$$\hat{Y} = f(X), \quad A \notin X \qquad \text{but} \qquad X \not\perp A \Rightarrow \hat{Y} \not\perp A$$

Seemingly neutral variable	What it reveals instead	How to check
Postcode	Race via residential segregation	Correlation with group labels
Occupation · school	Gender · socioeconomic background	Check SHAP contributions
App usage time · device	Family structure · income level	Gap change after removing the variable

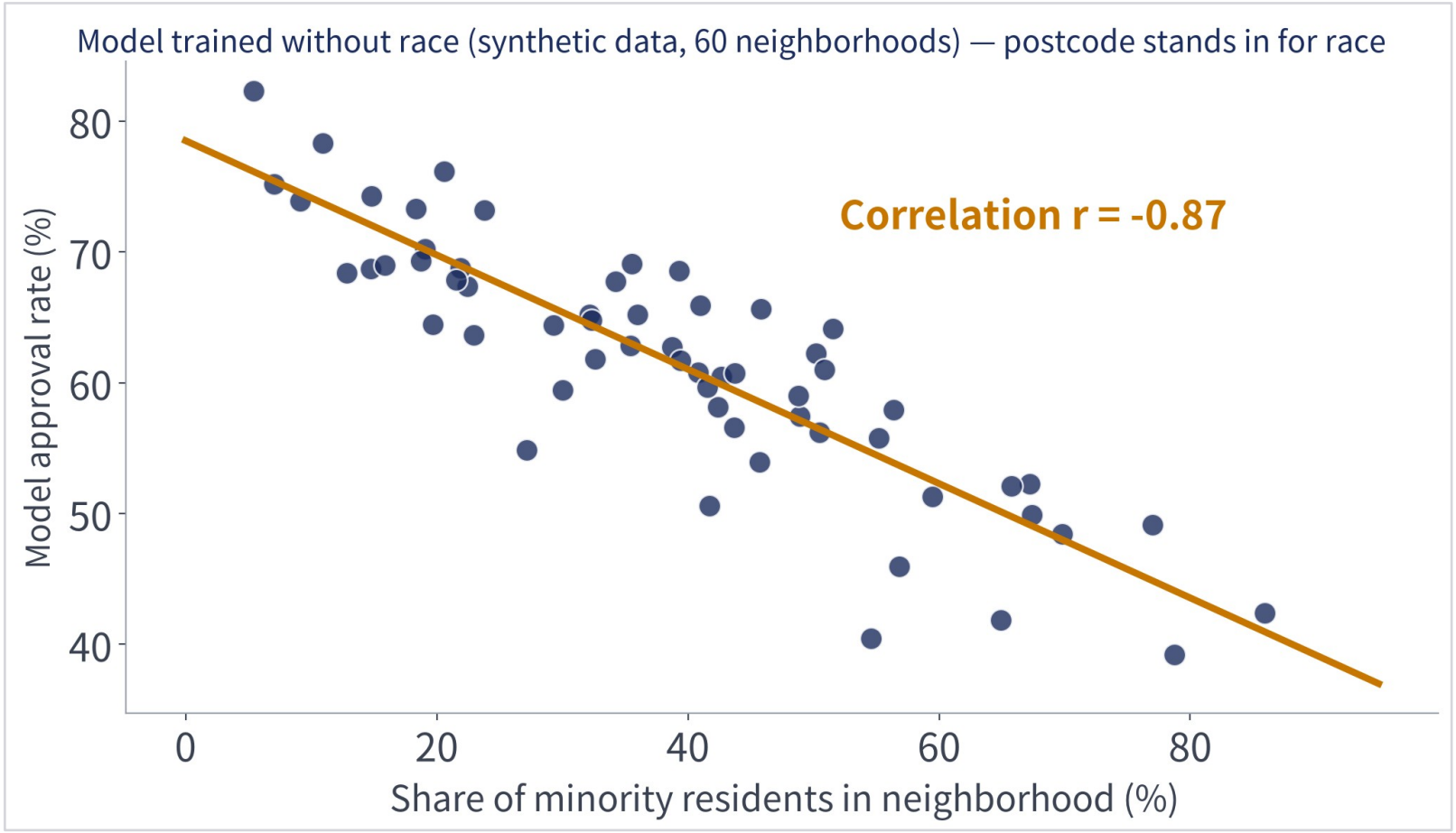
Test by function, not name. Ask not "is the column named race?" but "does the model behave like race?"

With postcode alone, approval rates track race

Neighborhood approval rates of a model trained without race (synthetic data, 60 neighborhoods)

How to read

- One dot = one neighborhood
- Further right = more minority
- Approval rate falls
- Deleting the variable fails



Demographic parity equalizes approval rates across groups

Demographic Parity (Statistical Parity) — ignores actual qualification Y

$$P(\hat{Y}=1 \mid A=\text{group } 0) = P(\hat{Y}=1 \mid A=\text{group } 1)$$

Source example: loan approval by gender

- 700 of 1,000 men approved (70%)
- 400 of 800 women approved (50%)
- Goal: bring both groups to 60%

Strengths and limits

- Usable without qualification data
- Fits quota and inclusion policies
- Ignores qualification gaps, inviting reverse-discrimination disputes

Equal opportunity gives the same chance to those who will repay

Equal Opportunity (Hardt · Price · Srebro 2016) — compares approval rates only among the qualified (Y=1)

$$P(\hat{Y} = 1 \mid \underbrace{Y = 1}_{\text{actual repayment}}, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

Source example: different numbers of qualified

- 700 qualified men, 500 qualified women
- 90% of the qualified approved in both groups
- Approvals differ: 630 vs 450

How it differs from demographic parity

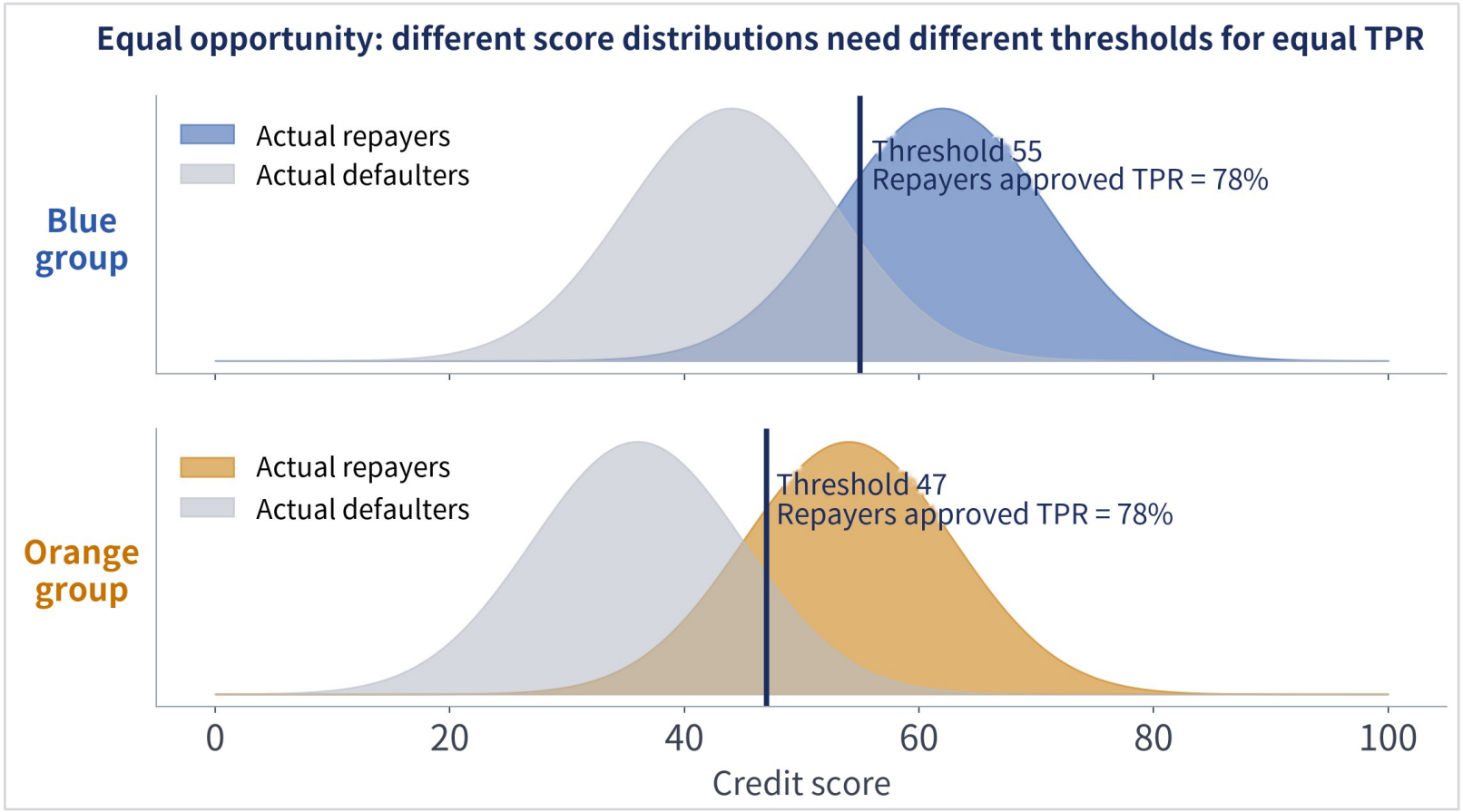
- Looks at the rate among the qualified, not the overall approval rate
- Small efficiency loss; the mainstream standard in finance
- Does not constrain false positives among the unqualified (FPR)

Thresholds must differ by group to equalize TPR

Distribution reproducing the source's loan-approval chart (Google 'Attacking Discrimination' simulation)

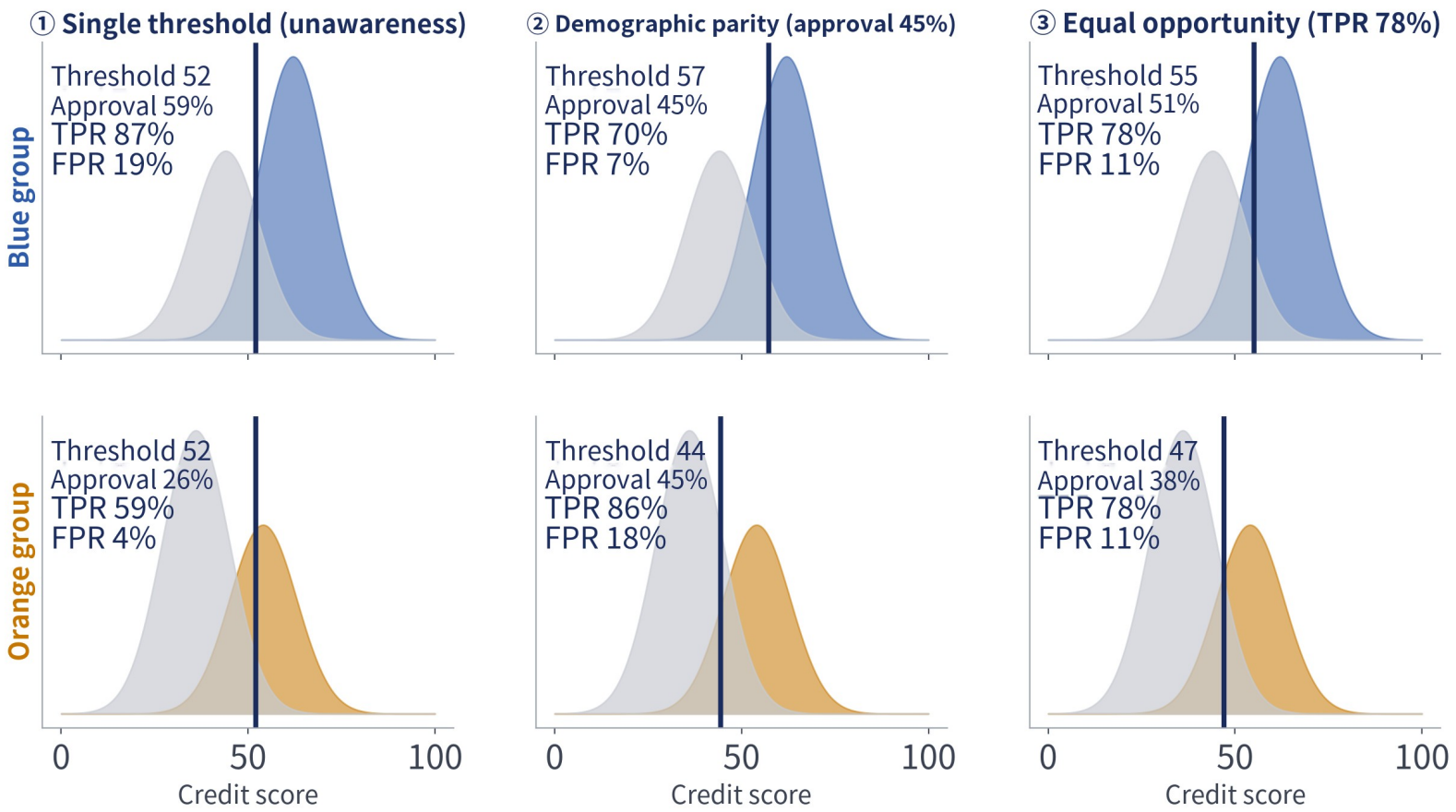
How to read

- Color: actual repayers
- Gray: actual defaulters
- Vertical line: threshold
- TPR 78% in both groups



Applying three policies to the same score distribution splits the results like this

A single threshold gives the orange group TPR 59%; demographic parity sets approval to 45%; equal opportunity sets TPR to 78%



Color = actual repayers, gray = actual defaulters (synthetic · repayment share: blue group 60%, orange group 40%)

Equalized odds requires the same error rates for everyone

Equalized Odds — equalizes both TPR and FPR; stricter than equal opportunity

$$P(\hat{Y}=1 \mid Y=y, A=0) = P(\hat{Y}=1 \mid Y=y, A=1), \quad y \in \{0,1\}$$

→ TPR$(y=1)$ and FPR$(y=0)$ are equal in both groups

Source example	Men (700 repay · 300 default)	Women (600 repay · 400 default)
TPR (repayers approved)	90% → 630	90% → 540
FPR (defaulters approved)	20% → 60	20% → 80
Overall approval rate	69%	62%

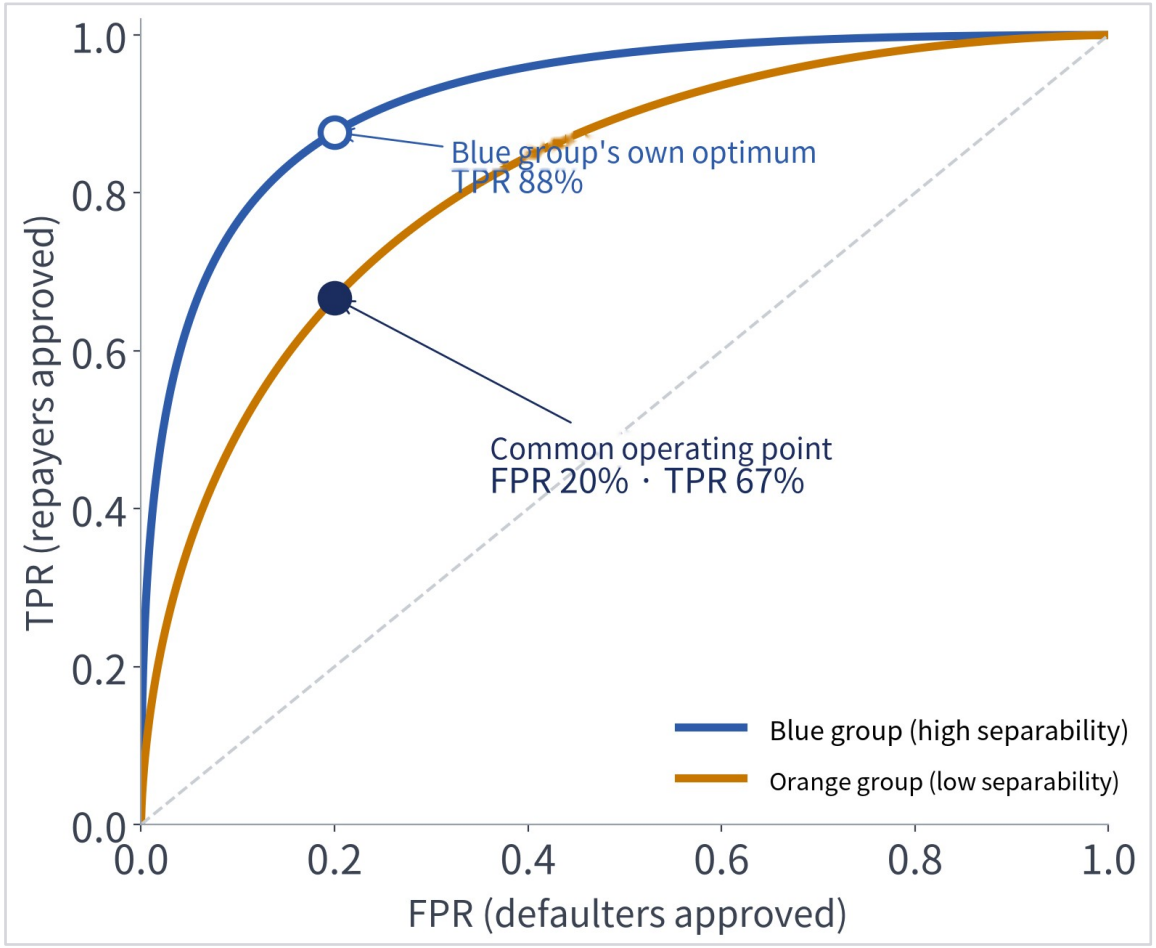
Equal error rates still give approval rates of 69% vs 62%. Equalized odds and demographic parity cannot hold together.

Equalized odds is only possible where the two ROC curves meet

Hardt et al. (2016) — with different separability, you must match the curve of the weaker group (synthetic distributions)

How to read

- Curve = full threshold range
- At the same FPR of 20%
- TPR 88% vs 67%
- Matching needs randomization



Predictive parity and calibration equalize the reliability of approvals

Predictive Parity · Calibration — the view of the decision-maker, not of those receiving decisions

$$\underbrace{P(Y=1 \mid \hat{Y}=1, A=0) = P(Y=1 \mid \hat{Y}=1, A=1)}_{\text{Predictive Parity: equal PPV}} \qquad \underbrace{P(Y=1 \mid S=s, A=a) = s}_{\text{Calibration}}$$

Meaning

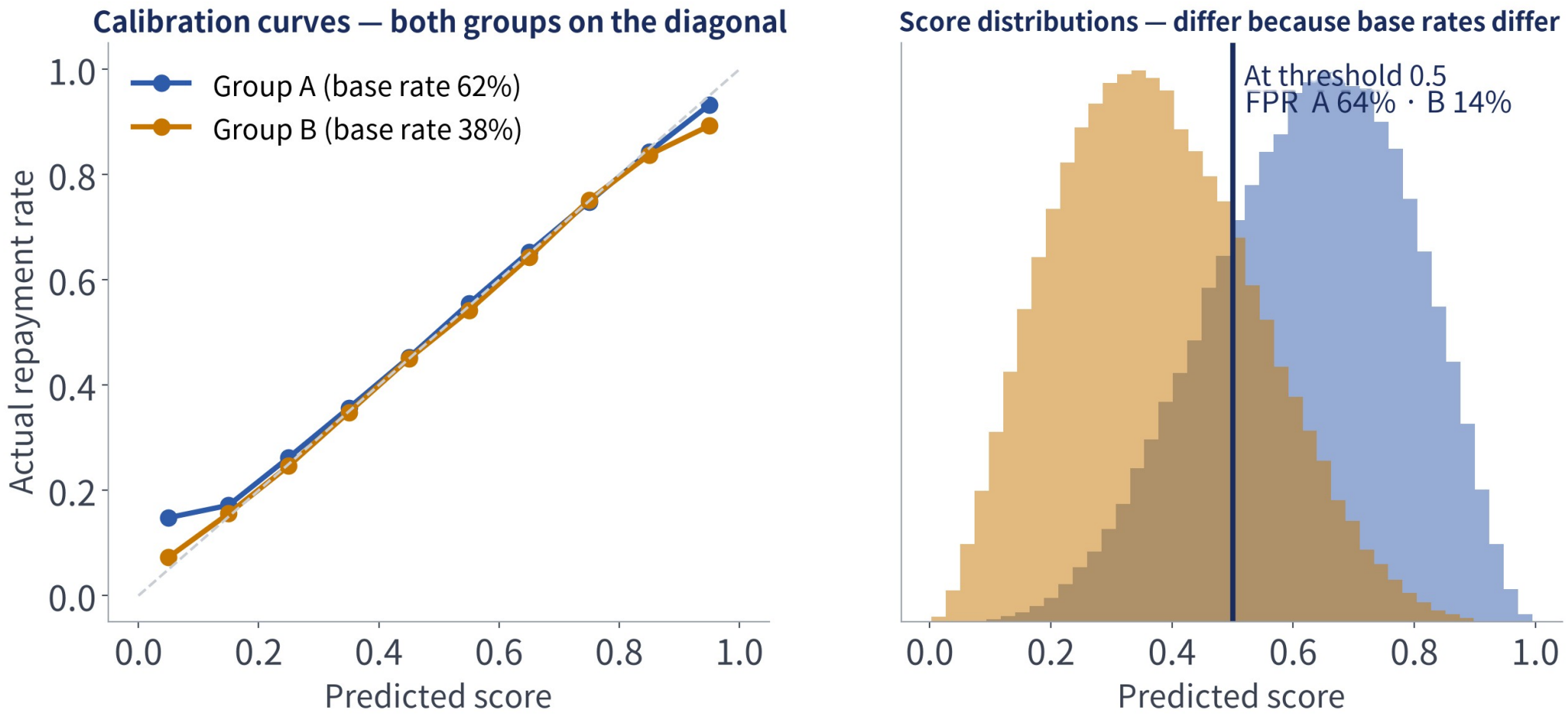
- Repayment probability of approved applicants is independent of group
- A score of 0.7 means 70% repay, whoever you are

Who prefers it

- Pricing and risk management units
- Fairness standard of the COMPAS developer

Even perfectly calibrated scores can have very different false-positive rates

Synthetic data — both groups' calibration curves lie on the diagonal, yet at threshold 0.5 FPR is 64% vs 14%



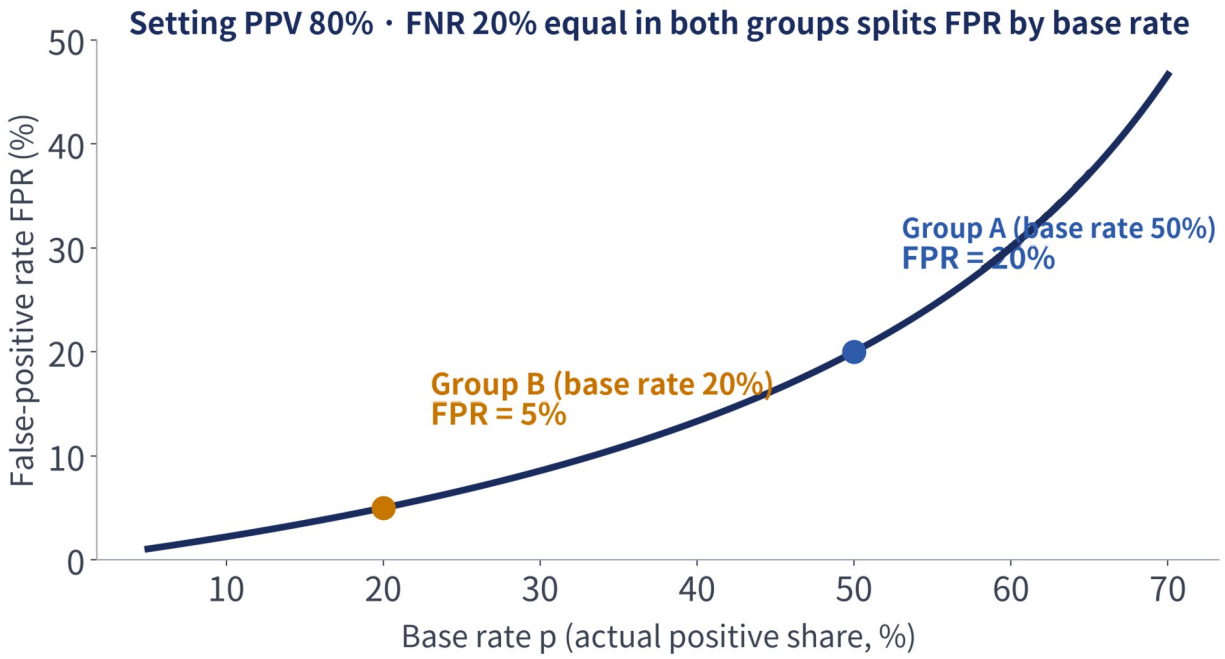
With different base rates, calibration and equal error rates cannot hold together

Chouldechova (2017) · Kleinberg et al. (2016) — base rate p is a group's actual positive share

$$\mathrm{FPR} = \underbrace{\frac{p}{1-p}}_{\text{base rate}} \cdot \frac{1-\mathrm{PPV}}{\mathrm{PPV}} \cdot (1-\mathrm{FNR})$$

What the formula says

- Equalizing PPV and FNR splits FPR
- All hold only when base rates are equal



Counterfactual fairness compares a person across imagined identities

Counterfactual Fairness (Kusner et al. 2017) — intervene on protected attribute A (A←a′) in a causal model

$$P\big(\hat{Y}_{A\leftarrow a}(U) = y \mid X = x, A = a\big) = P\big(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a\big) \quad \forall\, y,\, a'$$

Hold the same person (U), change only the protected attribute: the result must stay the same

Difference from equalized odds

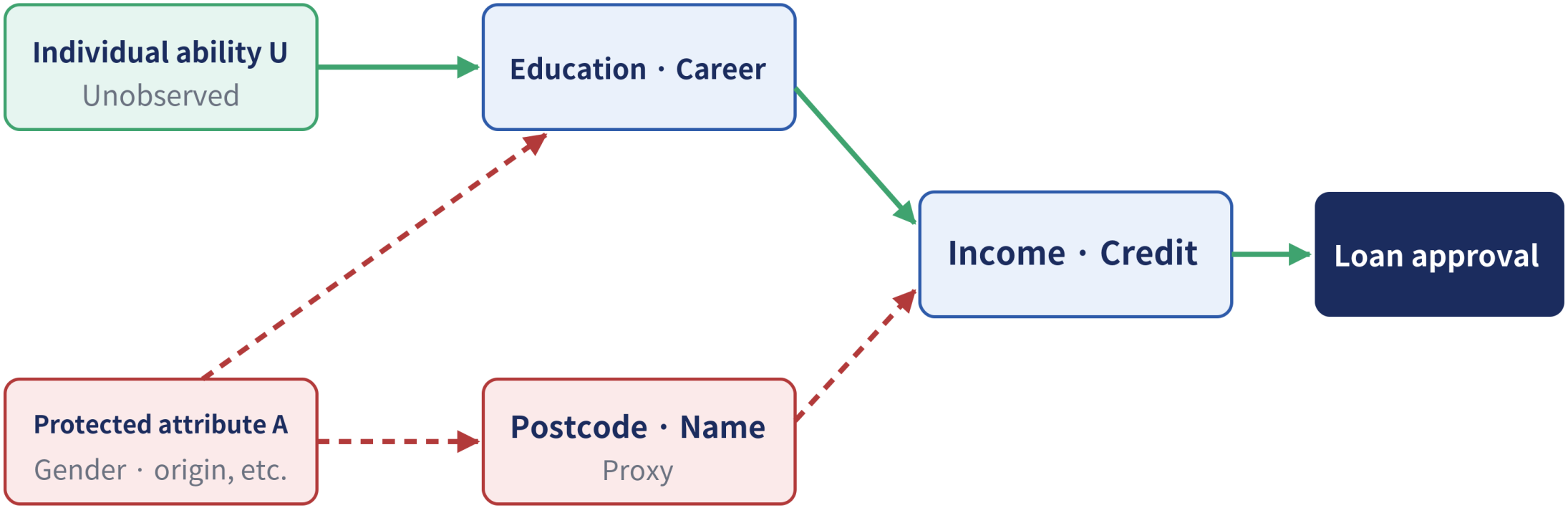
- Looks at individuals, not group averages
- A causal graph decides which paths are discriminatory

Implementation challenges

- Unobserved ability U must be estimated
- If the causal graph is wrong, so is the conclusion

A causal graph separates paths to keep from paths to cut

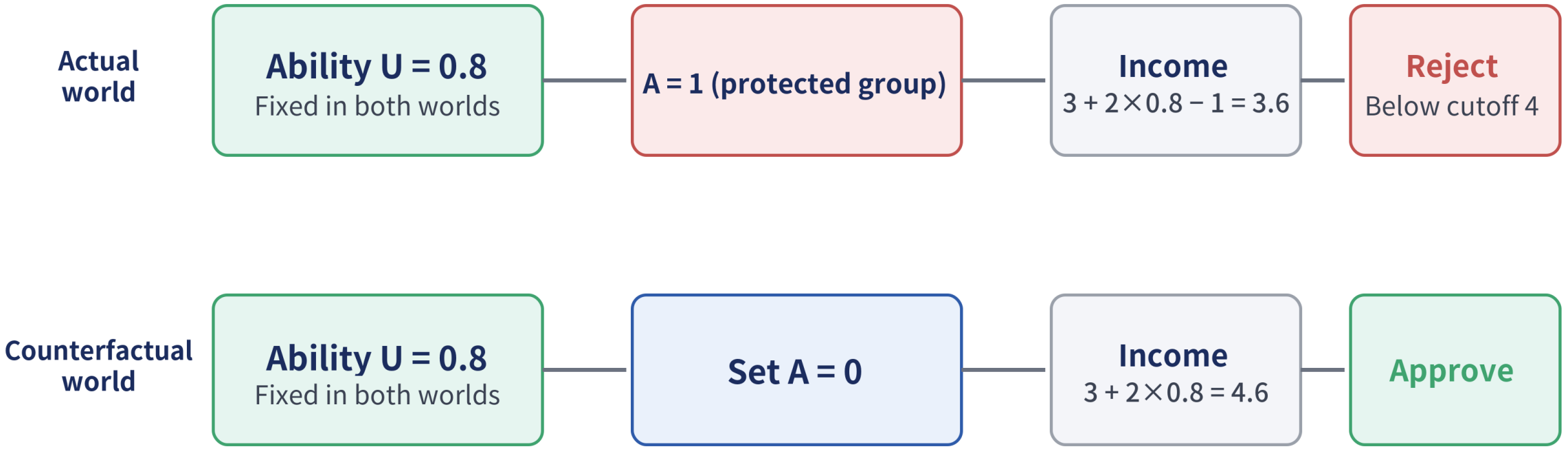
Keep the path from ability through income and credit to approval; block the path from the attribute through proxies



Green solid = paths to keep (ability → qualification); red dashed = paths to cut (attribute → proxy · blocked opportunity)

Counterfactual computation places the same person in two worlds

Toy model: income = 3 + 2U − A, approve if income ≥ 4 (calculation in Workbook 8)



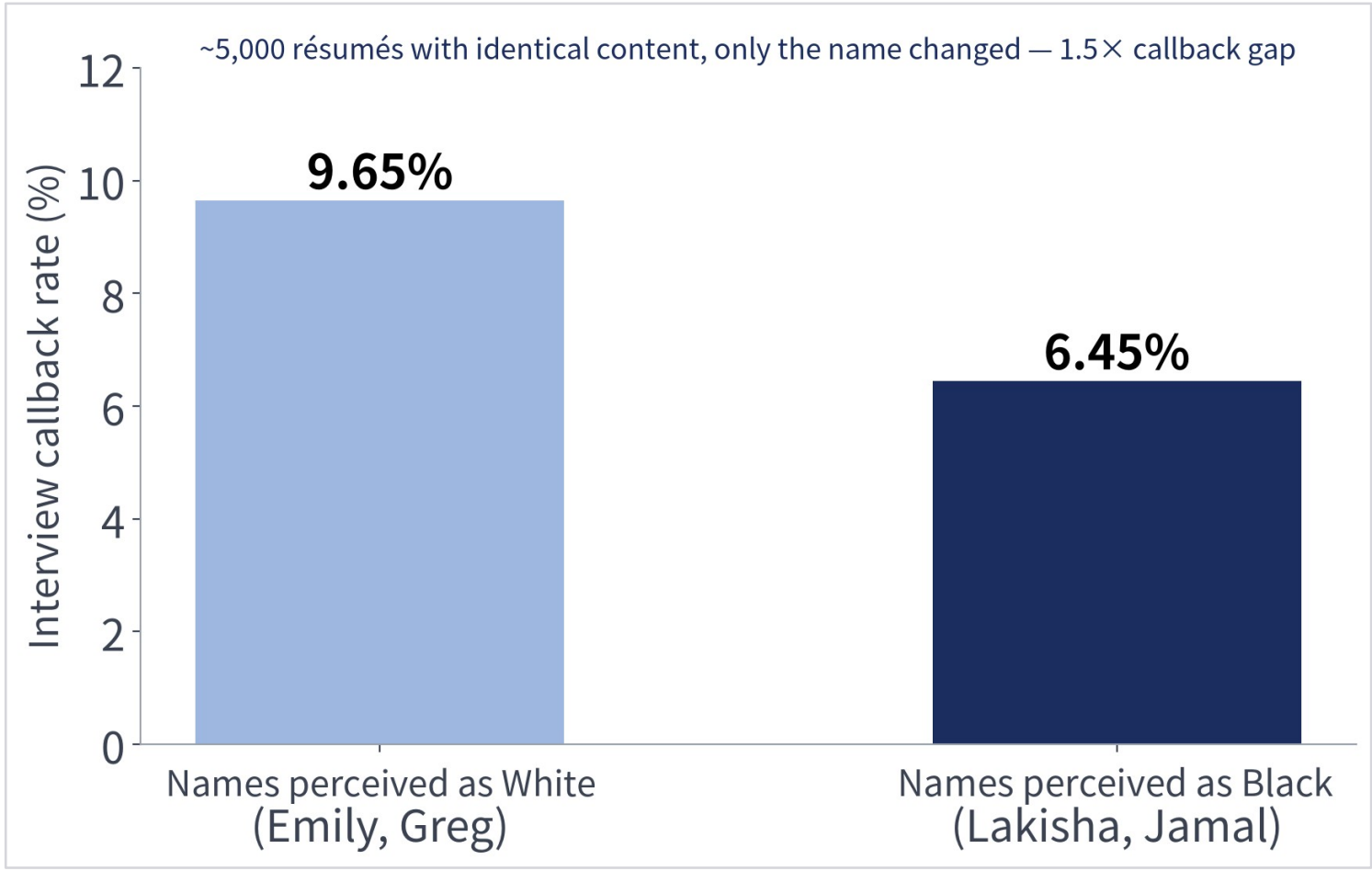
Same person (U fixed), only the attribute changed, and the decision flipped → counterfactually unfair, sent to human review

The name-swap résumé test is the counterfactual question in real life

Bertrand · Mullainathan (2004, AER) — fictitious résumés sent to 1,300+ job ads in Boston and Chicago

Teaching point

- Identical content
- Only the name changed
- Callbacks 9.65% vs 6.45%
- Audit AI the same way

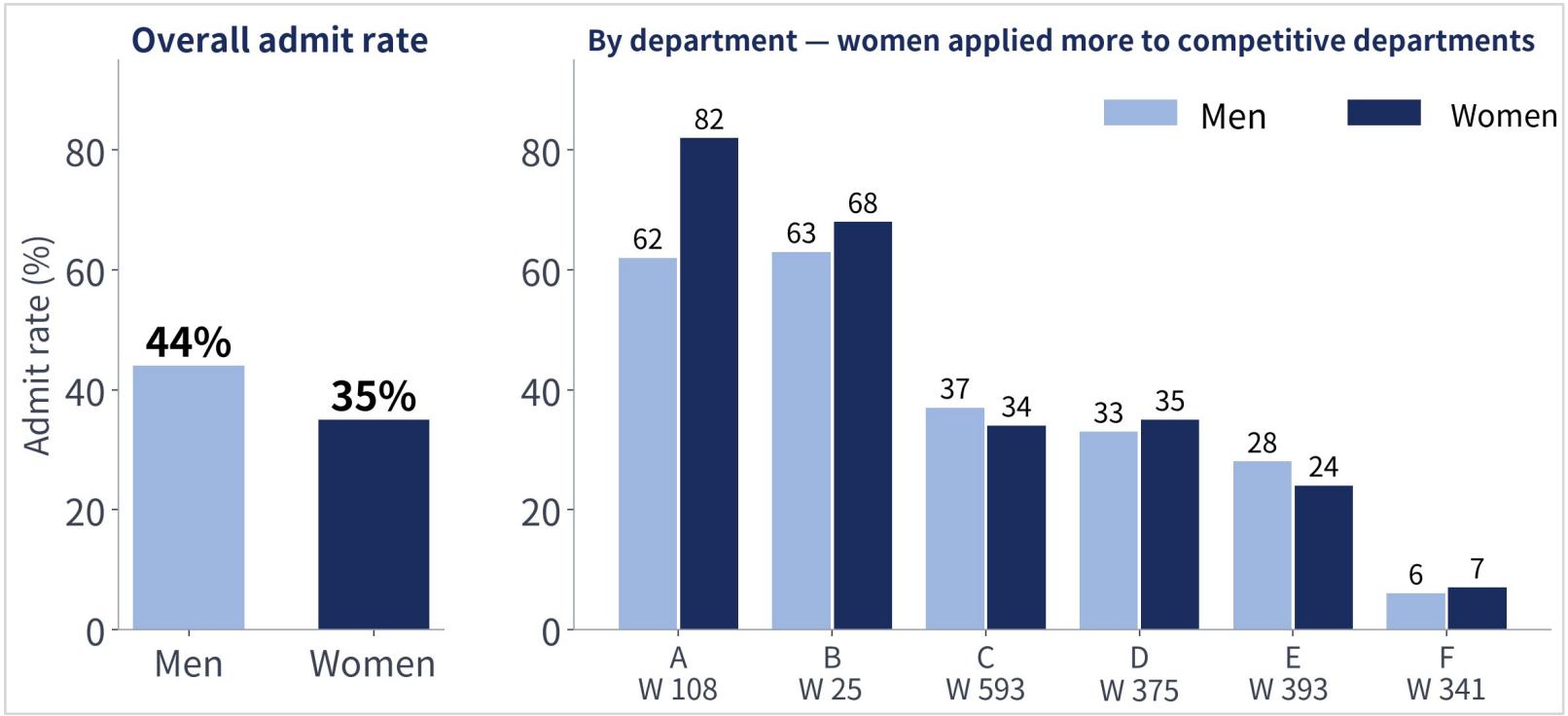


In Berkeley admissions, aggregation created the discrimination

Bickel et al. (1975, Science) — the source's '2021' corrected to the 1973 data

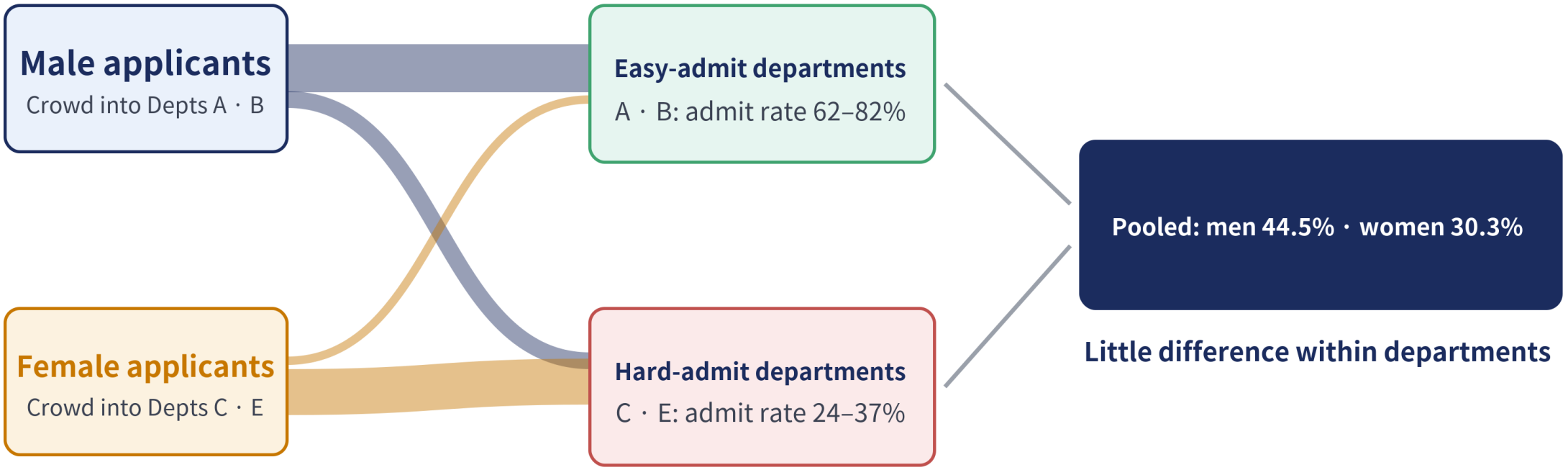
Counterfactual question

- Overall 44% vs 35%
- Negligible gaps by department
- Department choice is the cause
- Split the paths



Drawing the application paths reveals the cause of the aggregate gap

Women applying mostly to departments with low admission rates created the overall gap



Line width = number of applicants (Bickel et al. 1975, six major departments)

Product recommendation adds individual fairness and exposure fairness

Three metrics from the source's 'financial product recommendation model strategy' — two of individual, group and exposure fairness written as formulas

$$\underbrace{D\big(M(x), M(x')\big)}_{\text{outcome difference}} \leq L \cdot \underbrace{d(x, x')}_{\text{similarity of people}}$$

$$\text{Exposure}(G) = \frac{1}{|G|} \sum_{i \in G} \sum_{k} \frac{P(i \text{ at rank } k)}{\log_2(1+k)}, \qquad \frac{\text{Exposure}(G_0)}{\text{Utility}(G_0)} = \frac{\text{Exposure}(G_1)}{\text{Utility}(G_1)}$$

Individual fairness (Dwork 2012)

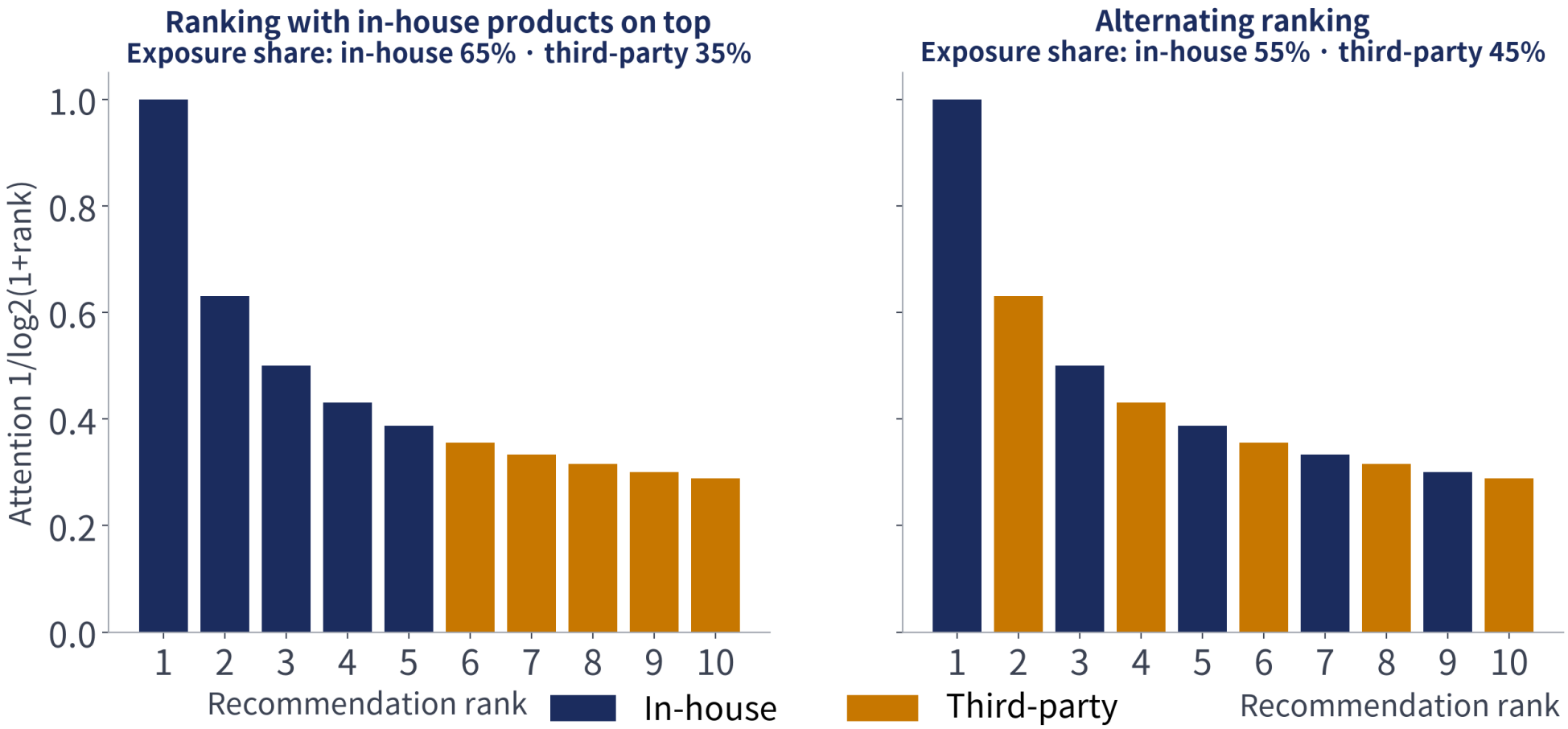
- Similar customers should receive similar recommendations
- The key dispute is who defines the similarity d

Exposure fairness (Singh · Joachims 2018)

- Divide top exposure in proportion to utility
- Catches bias that puts only in-house products on top

Taking the top five slots captures two-thirds of exposure

Five in-house and five third-party products — attention per rank set to 1/log2(1+rank), exposure shares computed



The harm you want to prevent decides which definition to use

The source's ROI figures by step could not be verified and were replaced with qualitative criteria

Harm to prevent	Primary definition	Financial example
Qualified people losing a chance	Equal opportunity	Loan and card approval
Wrongful high-risk labels	Equalized odds	Fraud detection · delinquency alerts
Scores meaning different things for different groups	Calibration · predictive parity	Premium and interest-rate pricing
A group being blocked from access altogether	Demographic parity	Allocation of inclusive-finance products
An individual judged differently because of an attribute	Counterfactual fairness	Large-loan review · appeals

03

PART 3 · Measurement and Mitigation

Measuring and Reducing Gaps

Metrics → group-level audit → pre-, in- and post-processing → cost of fairness

A practical audit starts with four gap metrics

DI = disparate impact ratio, SPD = approval-rate difference, EOD = TPR difference, AOD = average error-rate difference

$$\mathrm{DI} = \frac{P(\hat{Y}=1 \mid \text{protected})}{P(\hat{Y}=1 \mid \text{unprotected})} \geq 0.8 \qquad \mathrm{SPD} = P(\hat{Y}=1 \mid \text{protected}) - P(\hat{Y}=1 \mid \text{unprotected})$$

$$\mathrm{EOD} = \mathrm{TPR}_0 - \mathrm{TPR}_1 \qquad \mathrm{AOD} = \tfrac{1}{2}\left[(\mathrm{TPR}_0 - \mathrm{TPR}_1) + (\mathrm{FPR}_0 - \mathrm{FPR}_1)\right]$$

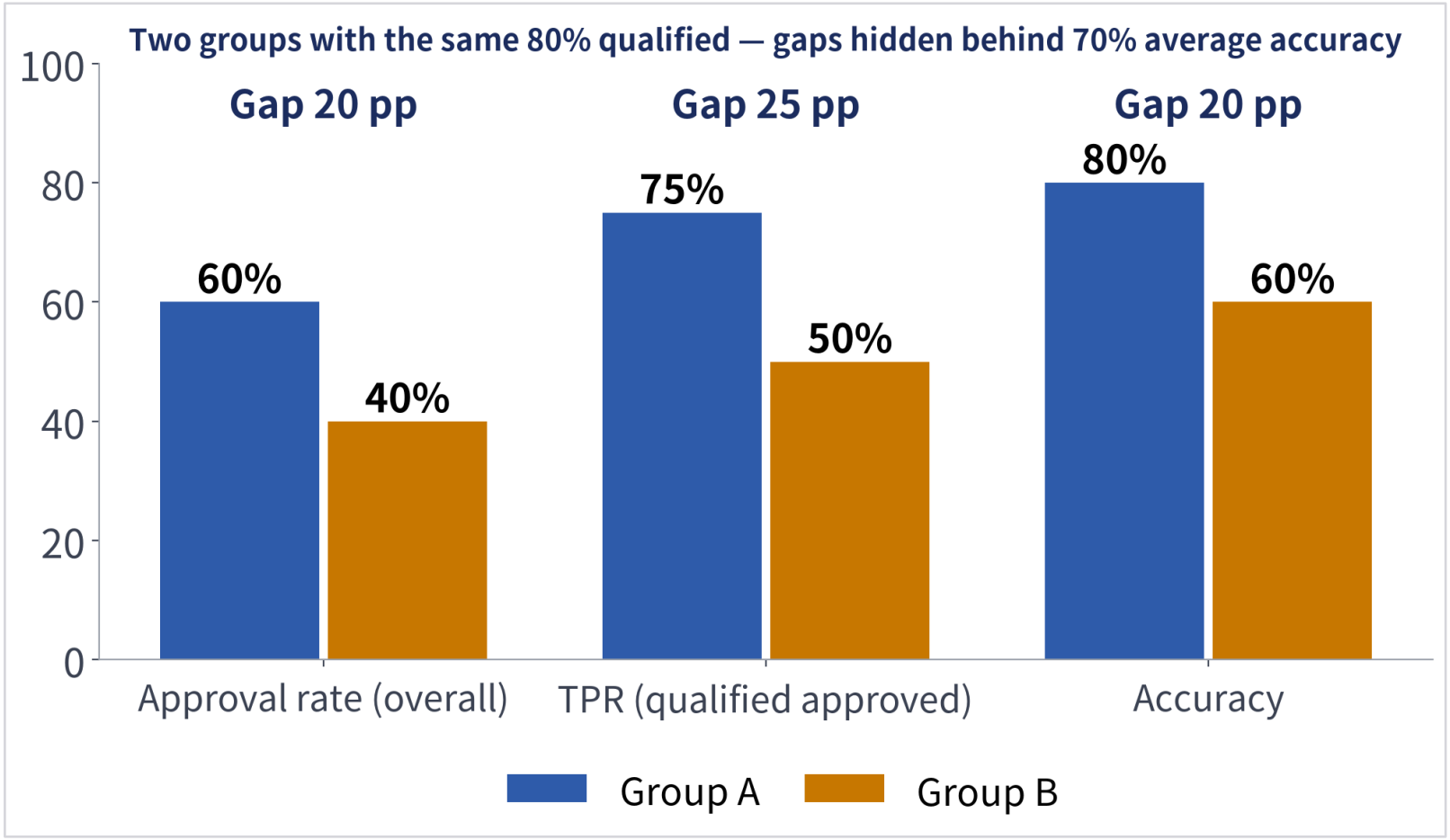
Metric	Common threshold	Applied to source example
DI (4/5 rule)	≥ 0.8	50% ÷ 70% = 0.71 → fails
SPD	|SPD| < 0.1	50% − 70% = −0.20
EOD	|EOD| < 0.05	Approval-rate gap among the qualified

A single overall accuracy hides gaps between groups

Two groups of 100, each with 80 qualified. Zero false positives, yet a 25 pp opportunity gap (calculation in Workbook 1)

How to read

- Same share qualified
- Approved 60 vs 40
- Overall accuracy 70%
- Gaps by group



Bias mitigation intervenes at three points: data, training and decision

Pre-processing · In-processing · Post-processing — techniques from the source's 8-hour curriculum

Pre-processing: $w_{a,\,y} = \dfrac{P(A=a)\,P(Y=y)}{P(A=a,\,Y=y)}$

Pre-processing (reweighting · resampling)

Leaves the model unchanged, but proxies may remain

In-processing: $\min_{\theta}\ \mathcal{L}_{\text{pred}}(\theta) + \lambda(|\Delta\text{TPR}| + |\Delta\text{FPR}|)$

In-processing (fairness regularization)

Tunes the accuracy-fairness balance, but needs λ selection and retraining

Post-processing: $\hat{Y} = \mathbf{1}\{s(x) > \tau_a\}, \quad \tau_a$ = group-specific threshold

Post-processing (group thresholds)

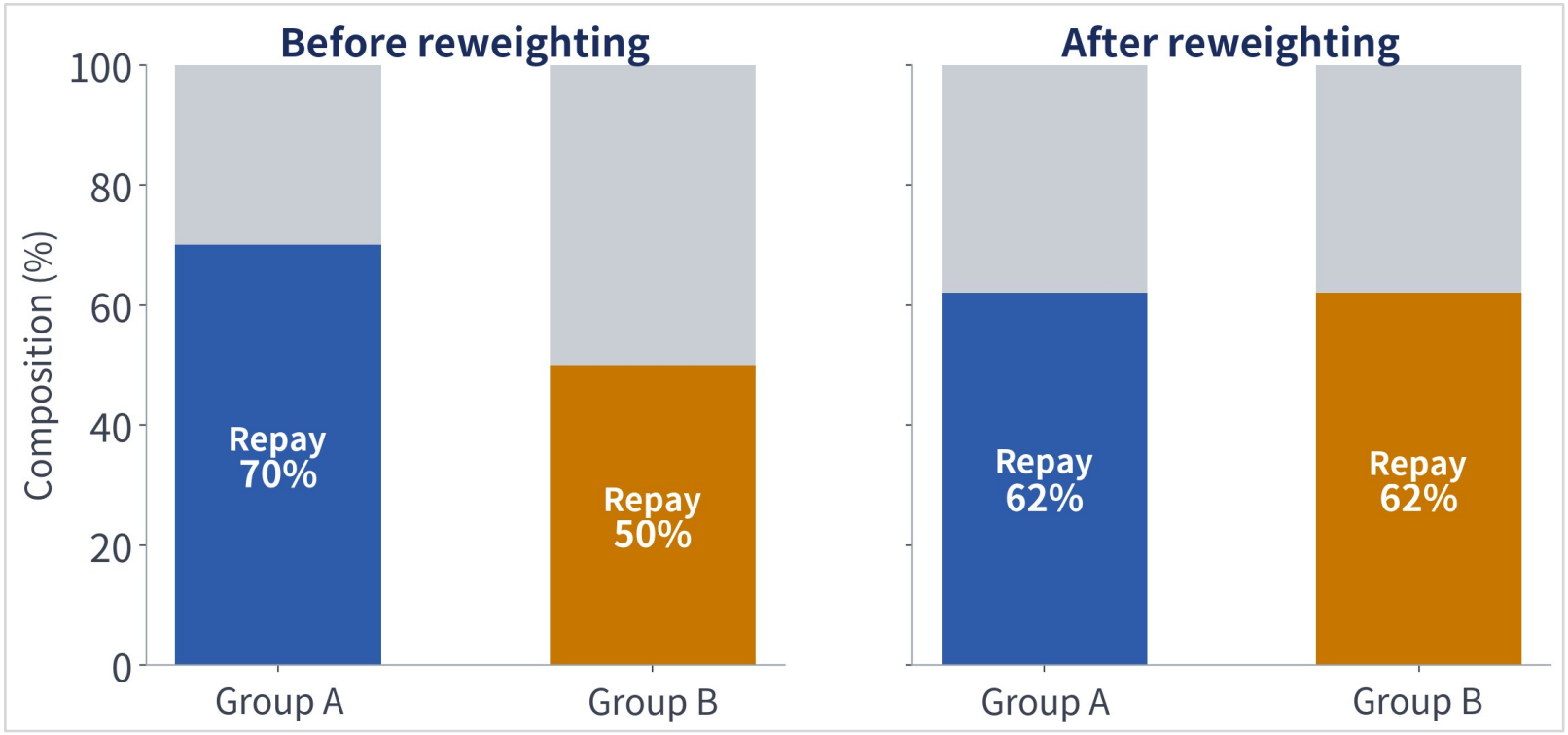
Applies immediately, but uses the attribute explicitly

After reweighting, both groups have the same repayment share

Weighting A 600 (420 repay) and B 400 (200 repay) brings both to 62% (calculation in Workbook 7)

How to read

- Color = repayment share
- Before: 70% vs 50%
- After: 62% vs 62%
- Head counts unchanged

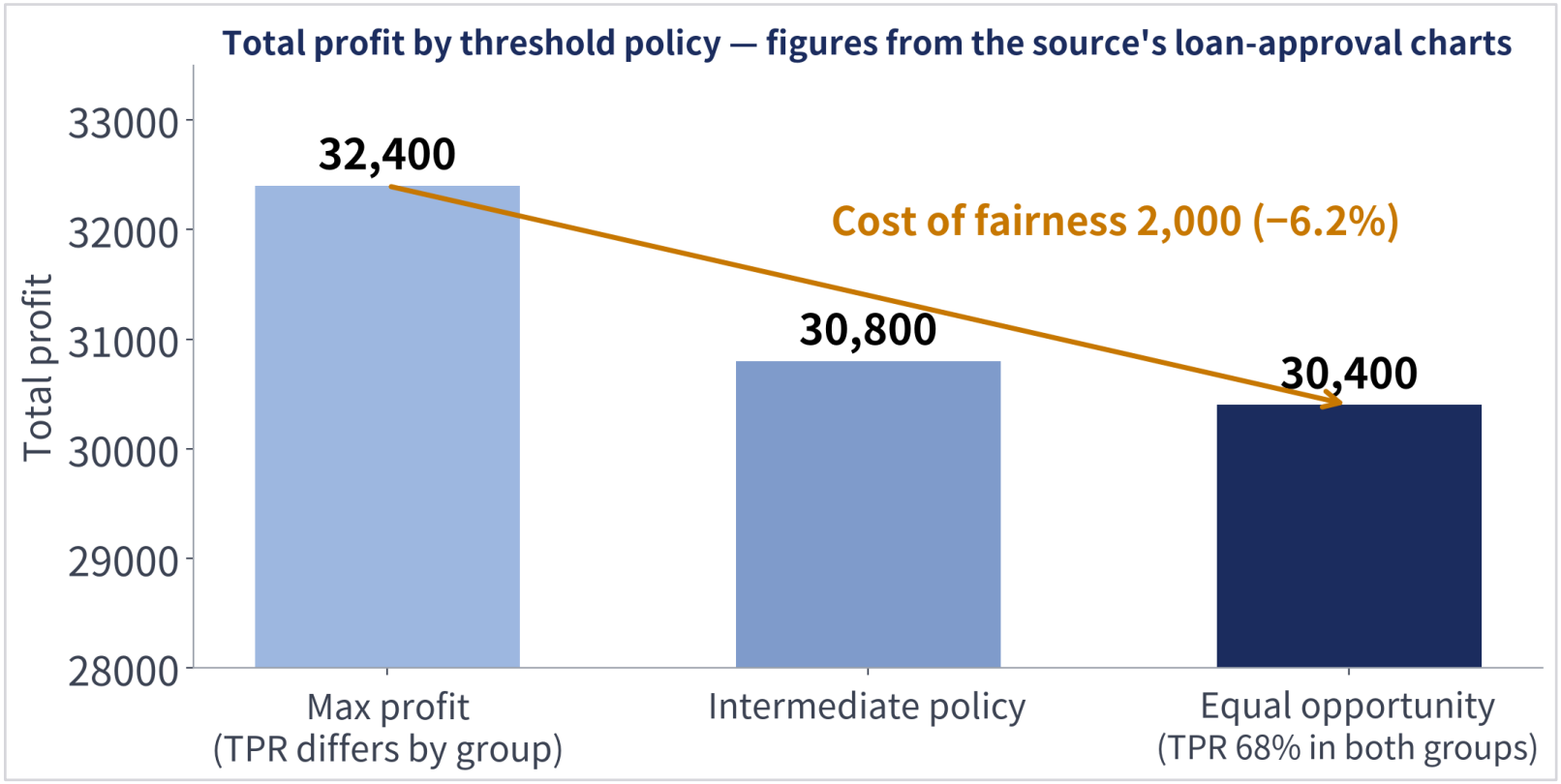


Equal opportunity cost about 6% of profit

Total profit figures from the source's three loan-approval charts — a trade-off where more fairness means less profit

Interpretation

- Max profit 32,400
- Equal opportunity 30,400
- Difference 2,000
- Compare with regulatory and reputational costs

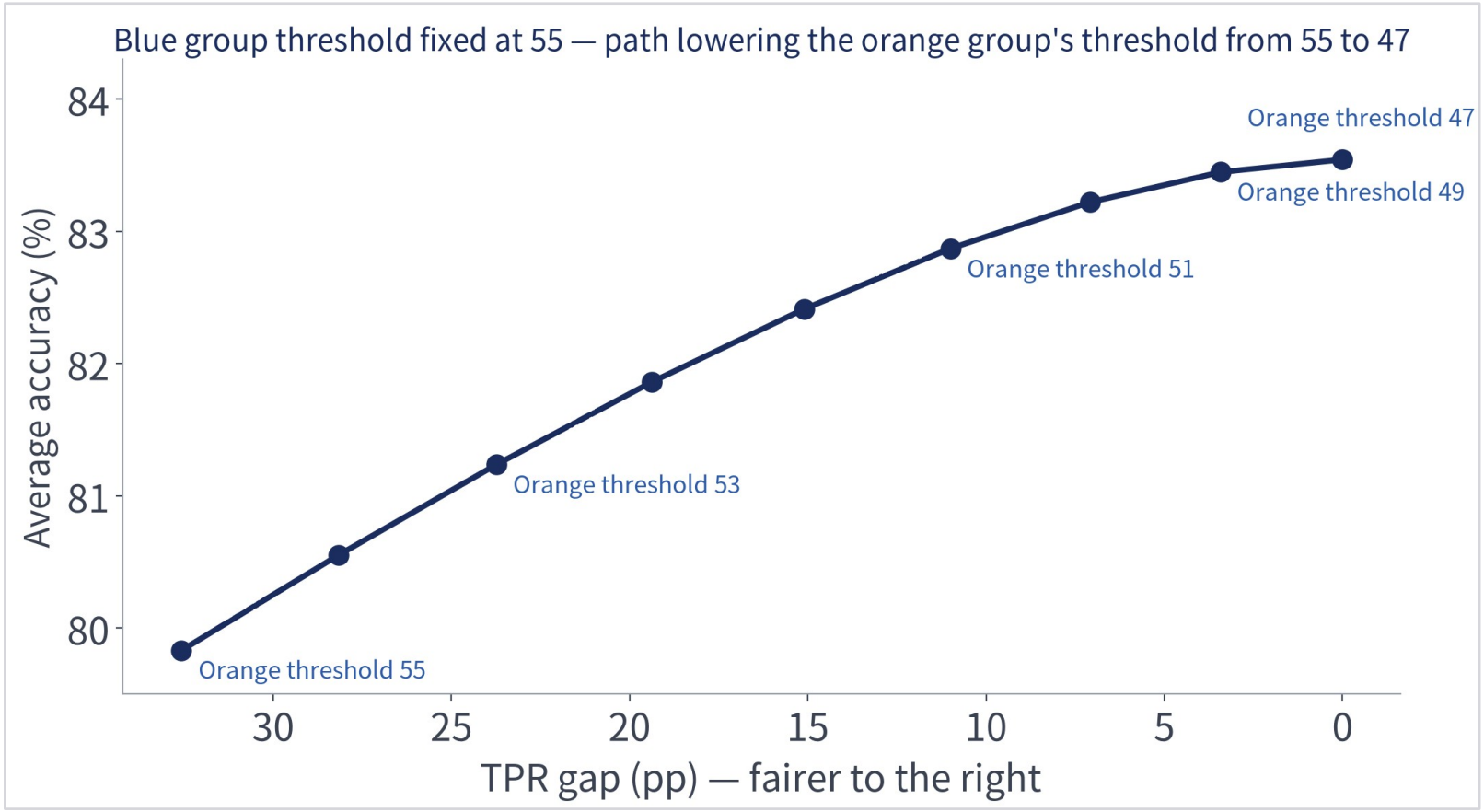


Fairness is not always costly

In the synthetic three-policy distribution, lowering the orange group's threshold shrank the TPR gap and raised accuracy

Interpretation

- Gap 33 pp → 0
- Accuracy 79.8 → 83.5%
- A single cutoff can be inefficient
- Verify cost empirically



04

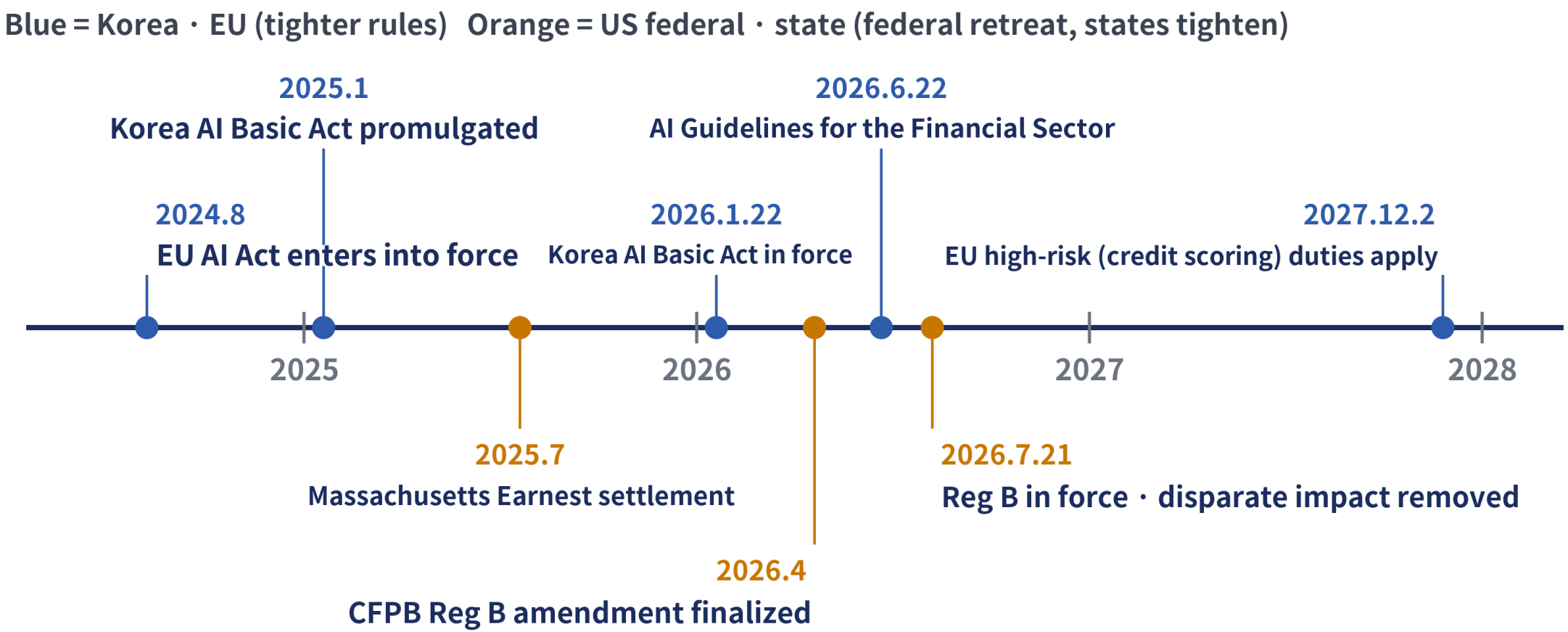
PART 4 · Governance and Regulation

Who Is Accountable and What Is Recorded

2026 regulatory landscape → diverging US trends → approval gates → impact assessment and monitoring

In 2026, regulation moved in opposite directions by region

Korea and the EU added record-keeping and explanation duties; the US federal level narrowed disparate-impact liability (as of October 2026)



Credit-scoring AI is high-risk in all three jurisdictions

The source's EU fine (€30 million · 6%) corrected to the final law

Jurisdiction	Key rule	Fairness-related obligations
EU	AI Act Annex III — credit scoring is high-risk	Data-bias management, documentation, human oversight; applies 2 Dec 2027
Korea	AI Basic Act (in force 22 Jan 2026) — loan screening is high-impact AI	Risk management, explanation and impact-assessment efforts; grace period of 1+ year
Korean finance	AI Guidelines for the Financial Sector (22 Jun 2026)	Seven principles; AI is an aid, final responsibility lies with staff
United States	CFPB Reg B amendment (21 Jul 2026)	Removes ECOA disparate-impact liability; covers intentional discrimination only

EU fines: up to €35 million or 7% of turnover for prohibited practices; up to €15 million or 3% for high-risk obligations.

The US federal level stepped back, but fair-lending risk remains

CFPB final rule, April 2026 — most of 64,500 comments opposed it, and lawsuits were filed

What changed federally

- Statistical outcome gaps alone are not an ECOA violation
- Only intentional discrimination (disparate treatment) is covered

What still remains

- For mortgages, the Fair Housing Act (FHA) recognizes disparate impact
- State laws in New Jersey, Massachusetts, California and others were strengthened

Industry advice converges on "Stay the course" — keep auditing

Put fairness checks at every gate of model approval

Combines the source's 8-hour-curriculum approval process with the three lines of defense (SR 11-7)

Ethics review

Before development

First ask who could be harmed

Fairness testing

Build · validate

Compare group TPR, FPR and DI with tolerances set in advance

Independent validation

Second line

Model risk management challenges proxies and assumptions

Approval · deployment

Board for high risk

Provide human-intervention paths and explanations of rejection reasons

Ongoing monitoring

3rd line · internal audit

Track gap metrics on a dashboard and decide on retraining

Group metrics and rationale logged at each gate enable regulatory response and internal audit

Impact assessment and rejection reasons are minimum conditions for trust

The source's seven AI impact-assessment questions and an adverse-action notice example

Seven impact-assessment questions

- Why is AI needed?
- What data, from where?
- How does the model work?
- What harm is possible?
- How is the risk reduced?
- Who monitors it?
- How can decisions be appealed?

Adverse-action notice example

- DTI of 45% exceeds the 40% limit
- Credit history is short (2 years)
- One recent delinquency
- Reduce debt and reapply in 6 months

Check three red flags before deployment

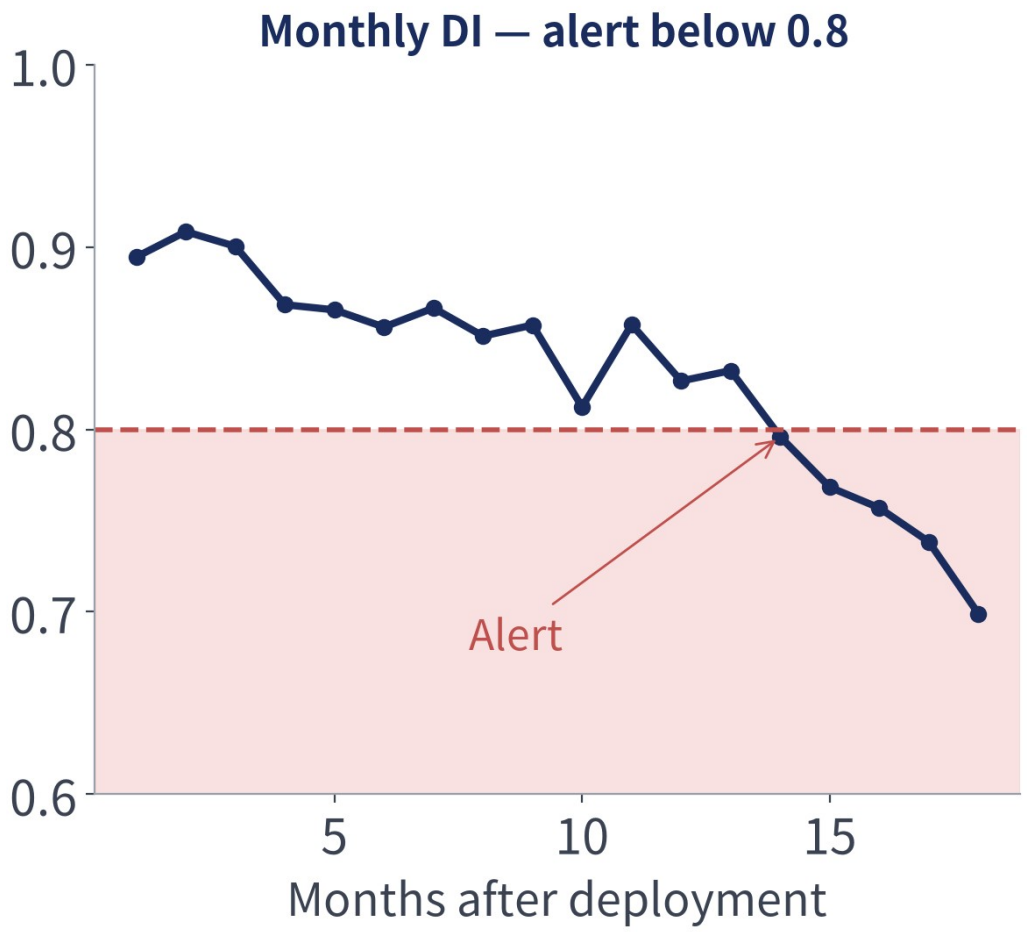
The deployment gate from FML Week 14 recast as a financial fairness checklist

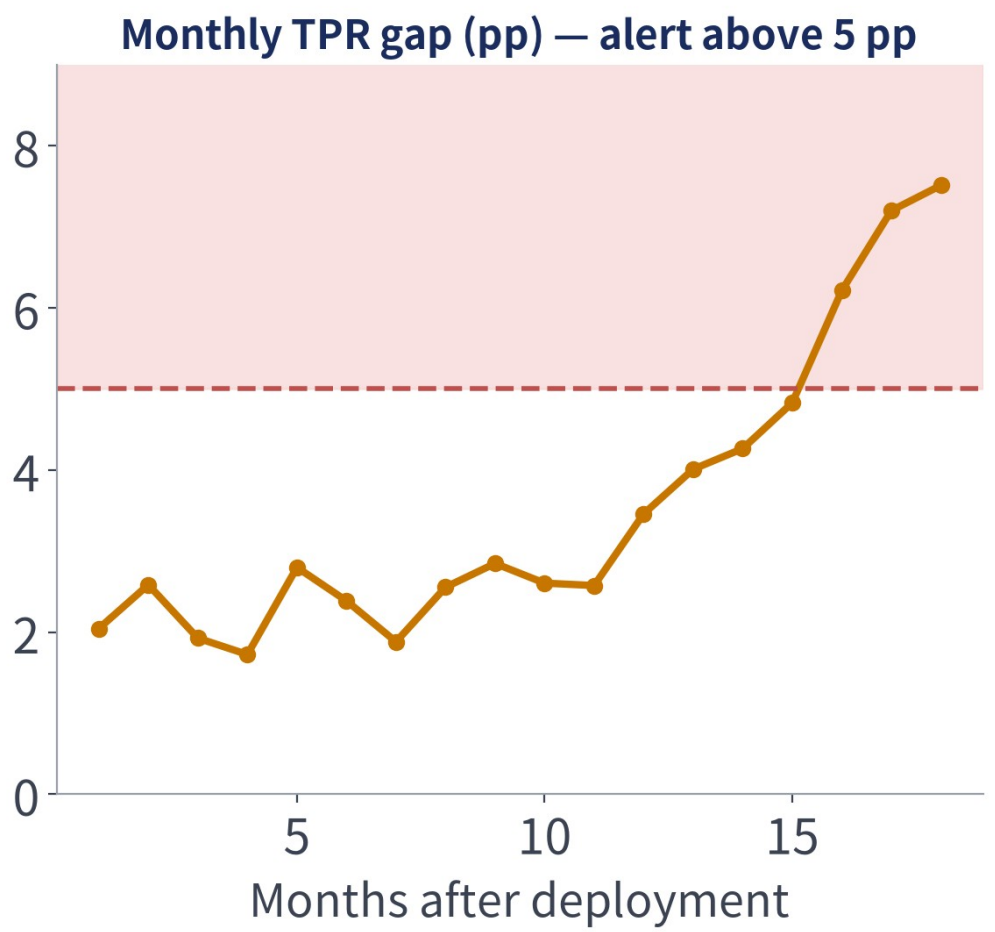
Red flag	What to check	Action if found
Unexplainable	Can SHAP explain the decision drivers?	Prune variables · compare with simple models
Gap exceeded	Are group TPR, FPR and DI within tolerance?	Revalidate after applying mitigation
Causality unclear	Was it checked whether proxies drive decisions?	Variable-removal tests · domain review
Distribution shift	Have group mix or base rates changed after deployment?	PSI alert · retraining

Always report both gaps (demographic parity and equal opportunity) in the model card, and set tolerances before deployment.

After deployment, track gap metrics monthly

Example data — as new customers changed the group mix, DI fell below 0.8 and the TPR gap rose above 5 pp





Discussion questions

Q1	If one credit model is used for two groups with different base rates, which definition should take priority?	**Choice of definition**
Q2	Without collecting protected attributes, how can gaps be measured?	**Measurement**
Q3	Why should Korean lenders keep auditing after the US removed disparate-impact liability?	**Regulation**
Q4	When an LLM chatbot recommends products, who should check exposure fairness?	**Governance**

Fairness is choosing a definition and recording it

Next session: compute each metric by hand in the workbook

1

Measure outcomes by group rather than deleting attributes

Proxies always exist. Audit approval rates, TPR and FPR, not the variable list.

2

Conflict between definitions is a mathematical fact

With different base rates, calibration and equal error rates cannot hold together. Decide the harm to prevent first.

3

Keep records and explanations even when regulation wavers

The EU and Korea expanded duties and US state laws tightened. Model cards and rejection reasons are the line of defense.

In one line: decide "whose errors to equalize," and document that choice

References

Category	Source
Definitions · theory	Hardt · Price · Srebro (2016) · Kusner et al. (2017) · Dwork et al. (2012) · Chouldechova (2017) · Kleinberg et al. (2016)
Cases	ProPublica (2016) COMPAS analysis · NYDFS (2021) Apple Card report · Bickel et al. (1975) · Reuters (2018) Amazon · Bertrand · Mullainathan (2004) résumé experiment
Recent research	Bowen · Price · Stein · Yang (2025) LLM mortgage screening · Singh · Joachims (2018) exposure fairness
Regulation	EU 2026/1744 (Digital Omnibus) · AI Basic Act (22 Jan 2026) · AI Guidelines for the Financial Sector (Jun 2026) · CFPB Reg B final rule (Apr 2026) · Massachusetts Attorney General (Jul 2025)